Installation and Configuration of WebSphere Liberty Profile for CCH Tagetik

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Introduction

This manual covers the procedures for installing and configuring the application on IBM WebSphere Liberty Profile and deploying CCH Tagetik 5 CPM on it. Please see the CCH Tagetik Compatibility matrices for compatibility with the service pack of CCH Tagetik 5 which you would like to deploy.

The intended audience of this manual is anyone who needs to understand and install CCH Tagetik on the JEE compliant Java application server IBM WebSphere Liberty. This manual describes the installation of the application as a single node using the WebSphere administrative console.

Readers should be familiar with how to operate and manage WAS Liberty as well have general IT skills such as how to install Windows services or how to set up an application on Linux.

This manual is organized so that the body of the manual contains all the steps necessary to complete a basic deployment of the application. The Appendices contain additional and more advanced information which are not covered in the main body of the manual. Please note that this manual is not intended to substitute the documentation provided by the developers and vendors of the AS for the installation of the AS.

The deployment of the application requires valid data sources and this manual assumes that the valid data sources are available. Instructions on how to create these data sources are provided in a specific manual available from CCH Tagetik.

Please contact CCH Tagetik concerning any questions or comments related to the contents of this document.

Overview

The installation of the application on the server tiers consists of two primary tasks:

- the creation of the application databases/schemas on a compatible RDBMS and,
- the configuration of the JEE application server followed by the deployment of the application JEE EAR archive onto a JEE compliant Application Server.

This installation manual focuses on the installation procedure for step II, and assumes that the application databases have already been created before testing the application deployment.

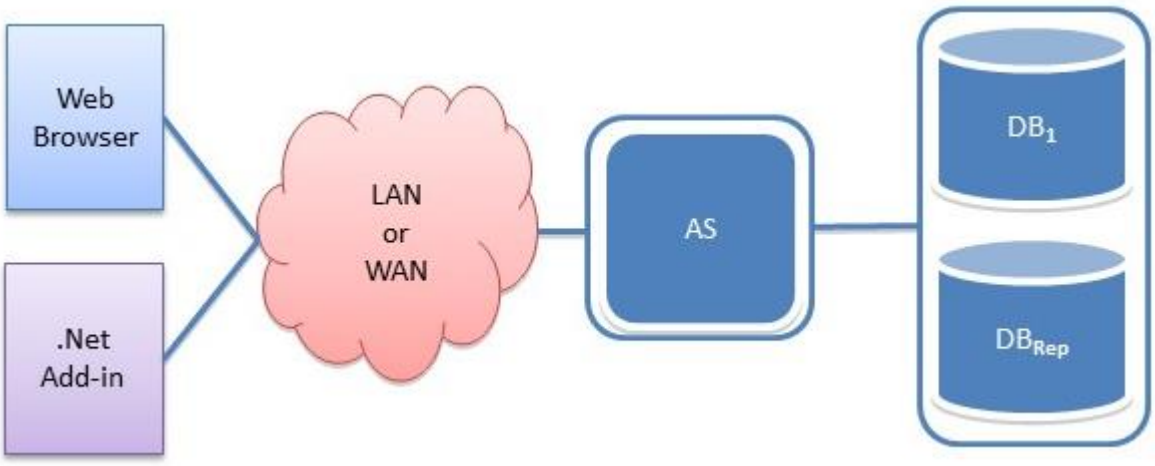


Figure 1 The basic architecture of the application.

The basic architecture of the application is shown in Figure 1. This manual covers the application server (AS) tier in the above figure. Step I of the complete installation procedure relates to the database server tier (DB).

Step II can be further divided into the following steps.

1. Check the prerequisites before the installation procedure.
2. Install and/or configure the JDK/JVM.
3. Configure the data sources which allow the AS to communicate with the RDBMS.
4. Configure, if necessary, the ports to be used by the AS interfaces and for the application.
5. Configure WebSphere and the application server logging.
6. Deploy the application EAR file on the AS.

Prerequisites

To install, configure, and deploy the application the following prerequisites must be satisfied. The following checklist is provided for the reader.

- **CPM Application Version**: Please check the compatibility matrices for compatibility with the Liberty Profile.
- **WAS Liberty Profile:** A compatible version of WebSphere Liberty should be installed. Please check the compatibility matrices for compatibility with the Liberty Profile.
- **Administration Web Console**: These instructions assume that the WAS web management console is available.
- **Installation User Privileges**: the person installing has access to a (local) administrative user on the application server hosting the AS. For Windows, a local admin user and for Linux, a “root” or a user with equivalent rights is required.
- **WAS Installation User Privileges**: The person installing the application on WAS should have administrative privileges, i.e. the ability to defined data sources, configure and deploy applications.
- **AS-DB Communication Open**: the AS host server should be able to communicate to the RDBMS host server. If there is a firewall between the two servers, the appropriate ports should be opened and traffic permitted depending on the configuration of the RDBMS.
- **JDK Installed**: A compatible version of a JDK must be installed on the application server host by a user with administrative privileges. Please check the compatibility matrices for compatibility with the Liberty Profile.
- **EAR & License Files**: the installation media must be available and readable to the WAS instance. In particular, the application EAR file, and the application license file.
- **DBs Installed**: At least the minimum 2 databases/schemas, a Repository and a business database/schema should be created with the necessary access configurations on a compatible RDBMS. This part of the installation procedure is covered in a separate manual.
- **Adequate AS Host Resources**: Tagetik requires a minimum of 8 GB of RAM on the host AS platform in order to deploy the EAR file. In general, customer projects require more than the minimum to provide adequate user response times. If you have not yet received an estimate of the recommended resources for the project, please contact CCH Tagetik or the project manager to request one. Alternatively, the CCH Tagetik Hardware and Software Architecture Overview contains an example of a minimal resource estimate for a limited number of users, i.e. 1-5 users in a relatively simple project.

JVM Settings

The installation procedure begins by configuring the JVM.
In order to provide JVM options to the server a file named `jvm.options` must be created under the `${server.config.dir}` folder. The following is an example of file content:

```
-Xmx8192m

# Log4j configuration system property.
-Dlog4j.configurationFile=file:/tagetik/config/log4j2.xml
# Import/Export Folder Definition
-Dtagetik.cpm.impexp.writableFolder=/tagetik/imp-exp
# Add Tagetik system properties below
```

Server feature Settings

This section describes how create a server and configure the respective server file:

1. Execute the command `server create ${serverName}` under bin folder
2. Edit the server.xml file under the `${server.config.dir}/wlp/usr/servers/${serverName}` path adding the following feature:

```
<featureManager>
    <feature>javaee-8.0</feature>
    ……
</featureManager>
```

   If you are configuring JakartaEE version:

```
<featureManager>
    <feature>jakartaee-10.0</feature>
    ……
</featureManager>
```

3. If not present, install the javaee-8.0 under the bin folder with the command `installUtility install javaee-8.0` (`installUtility install jakartaee-10.0` for JakartaEE version)

Configuring Data Sources

This section explains how to configure data sources inside WebSphere Liberty using the Administration Center.

Configuring shared libraries for JDBC drivers

In order to configure JDBC drivers, a shared library for each driver must be added to the server.
For each JDBC driver to be added, the following steps must be performed:

1. From the Administration Center home page open the “Server Config” section

2. Click on the server.xml link
3. In Design mode select “Server”
4. Click on “Add child”
5. Choose “Shared Library”
6. Choose an “ID” for this shared library (this will be used on step 10 in the next section)
7. Click on “Add child”
8. Choose “Fileset”
9. Put into the “Base directory” field the directory on server containing the driver library
10. Put into the “Includes pattern” field the name or a pattern that identifies the java library

Configuring Data Source Parameters

For each data source to be added the following steps must be performed:

1. From the Administration Center home page open the “Server Config” section
2. Click on the server.xml link
3. In Design mode select “Server”
4. Click on “Add child”
5. Select “Data Source”
6. Fill with the correct JNDI name both the “ID” and the “JNDI name” fields
    a. For REPOSITORY datasource use “`jdbc/DBEASYFINANCE2_REPOSITORY`”
    b. For applicative data sources use the “`jdbc/DBEASYFINANCE2_D00[0-23]_MAIN`” format (e.g. “`jdbc/DBEASYFINANCE2_D0000_MAIN`”, “`jdbc/DBEASYFINANCE2_D0001_MAIN`”, etc.)
    c. For secondary data sources use the “`jdbc/DBEASYFINANCE2_SECONDARY_00[0-25]`” format (e.g. “`jdbc/DBEASYFINANCE2_SECONDARY_0000`”, “`jdbc/DBEASYFINANCE2_SECONDARY_0001`”, etc.)
7. Select the data source entry in the left pane
8. Click on “Add child”
9. Choose “JDBC Driver”
10. Add a reference to the driver shared library into the “Shared library reference” field (as defined in step 6 in previous section)
11. Select the Data source entry in the left pane
12. Click on “Add child”
13. Choose “Connection Manager” and properly configure connection settings.
14. Select the data source entry in the left pane
15. Click on “Add child”.
16. Choose “Default Container Managed Authentication Data”
17. Configure username and password for this data source.
18. Select the data source entry in the left pane and properly configure the connection properties for the current DBMS.

Deploying CCH Tagetik Application

The application .ear file must be copied into the `${server.config.dir}/dropins` folder (e.g. `/opt/wlp/usr/servers/myserver/dropins/`). The application will be automatically deployed once the copy is finished.
Installed applications can be monitored from the Administration center at `https://{server}:{port}/adminCenter/#explore`.

Logging Configuration

Application logging can be enabled and configured adding the following system property inside the `jvm.options` file:
`-Dlog4j.configurationFile=`*`[path_to_log4j2.xml]`*

The above system property must contain a path to a valid log4j2.xml configuration file.

Glossary

The following table provides a short glossary for the terminology used in this manual.

AS	Application Server
DB	Database
RDBMS	Relational Database Management System, i.e. MS SQL Server, Oracle, PostgreSQL
JEE	Java/Jakarta Platform Enterprise Edition
EAR	JEE Enterprise Archive, a file format used in JEE enterprise applications.

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